# Gutenberg Atlas: Publication-Year and Author Metadata for English-Language Project Gutenberg Books

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## Abstract

Gutenberg Atlas is a book-level metadata dataset for English-language Project Gutenberg. For 21,296 books, it provides publication years derived from HathiTrust bibliographic matches, author lifespans, Wikidata gender and citizenship claims, and stable identifiers linking the source records. Records were linked deterministically to Wikidata authors and matched to HathiTrust bibliographic holdings. In a stratified sample of 199 release records, the derived years were within one year of sourced reference years for 87.4% (95% CI 82.7–92.1). The dataset supports historical corpus construction, digital humanities, bibliographic research, and evaluation of metadata-linkage methods.

**Keywords:** Project Gutenberg; HathiTrust; Wikidata; bibliographic metadata; digital humanities; historical corpora

## 1. Overview

Project Gutenberg is a large, continuously updated collection of machine-readable books, but its catalog does not systematically record when a work first appeared in print (Project Gutenberg, n.d.). Existing releases provide only part of the metadata needed for historical research: PG-19 includes publication dates but targets language modeling (Rae et al., 2019), whereas the Standardized Project Gutenberg Corpus includes author lifespans but not book-level first-publication years (Gerlach & Font-Clos, 2020). Related work has enriched German-language corpora with publication and demographic metadata (Rohrbacher, 2025; Rohrbacher & Schrittesser, 2026), while NovelTM documents HathiTrust metadata for English-language fiction (Underwood et al., 2020).

Gutenberg Atlas combines book-level publication-year estimates, structured author metadata, and sourced date validation for English Project Gutenberg in one reproducible build. Wikidata work dates served only as a cross-check because just 41 validation books matched a dated work. No book texts are redistributed. The resource is intended for studies that need temporal and author covariates on the same books without obscuring which archive supplied each value.

### 1.1 Repository location

**Repository:** https://github.com/jaytimm/gutenberg-atlas
**Dataset DOI:** https://doi.org/10.5281/zenodo.23628668

The repository contains the frozen release, retained candidate records with decision codes, validation records, documentation, build scripts, and Supplementary File 1. The DOI identifies the archived version 1.0.0 release (Timm, 2026).

### 1.2 Context

Project Gutenberg supplies catalog identity, HathiTrust bibliographic evidence, and Wikidata structured author claims. Source fields are retained so values remain traceable and reusable with Project Gutenberg texts. The release also preserves excluded candidates and decision codes, allowing users to audit coverage and apply different inclusion rules rather than treating the released corpus as a neutral sample of publishing history.

## 2. Method

### 2.1 Steps

The build began with a frozen Project Gutenberg RDF catalog: 61,865 English-only text records (Project Gutenberg, n.d.). Restriction to one named personal author and no other contributor left 38,550 eligible books, excluding anonymous, multi-author, and most translated or edited works. The monthly HathiTrust extract `hathi_full_20261001` was filtered to English books (HathiTrust Digital Library, 2026a). Normalized titles were matched within author surname, first exactly and then by prefix containment. The earliest imprint in the best tier became the year estimate. It is the earliest matched English-language HathiTrust edition, not necessarily first publication; translations, collections, and multi-volume sets require particular caution. For a translation it may approximate the earliest matched English edition, while a volume can inherit the date of a collected or multi-volume set.

Project Gutenberg agent identifiers were linked through Wikidata property P1938, retaining human entities and their identifier, label, gender (P21), citizenship (P27), birth (P569), and death (P570) claims. These are dated source claims, not inferences from names or text (Vrandečić & Krötzsch, 2014). Project Gutenberg and Wikidata lifespans remain separate; conflicts are flagged.

A released book required a HathiTrust match, a Wikidata human link, a year of 1700 or later, and an estimate between birth plus 15 and death plus five. Project Gutenberg supplied lifespans, with Wikidata filling gaps and birth plus 90 used when death was missing. Other eligible records remain in `unverified_candidates.csv` with exclusion reasons.

Of 38,550 eligible books, 21,296 (55.2%) were released. Main exclusions were no HathiTrust match (9,535), no Wikidata link (5,208), and an estimate more than five years after death (2,251). Exclusion varied by author era (Supplementary File 1).

### 2.2 Sampling strategy

Publication-year accuracy was evaluated on 200 release books stratified by five periods and two match tiers. At least eight books represented each cell; remaining positions followed cell size. Sampling was without replacement (seed 20261021); Supplementary File 1 gives the allocation.

### 2.3 Quality control

Reference years were manually researched and stored with source URLs and notes. Years were resolved for 199 books; one composite edition remained unresolved. Accuracy meant agreement within one year. Cell estimates were weighted to the release, with a stratified normal 95% confidence interval. Sources included catalogues, digitized editions, reference works, and trade listings. For 23 books the reference was a separate HathiTrust record and thus not source-independent; excluding them gave 87.4% accuracy (95% CI 82.3–92.5). Other source restrictions changed accuracy by at most 1.2 points (Supplementary File 1). Each reference note records what the cited year represents, so users can distinguish evidence for first publication from dates attached to particular editions. This distinction matters when dates anchor narrow historical comparisons.

Weighted release accuracy was 87.4% (95% CI 82.7–92.1). Among 16 sampled books dated 1925 or later, 10 were correct. This sparse stratum represents a small share of the release and is reported as a limitation, not used as an upper date boundary. Of 28 errors, 25 estimates were late (median +7 years), usually reflecting a later edition or reprint. Validation measured accuracy but did not alter release rows. Because screening uses the estimated year, results apply only to released books, not the 17,254 excluded candidates; genuine posthumous editions may also be excluded.

**Table 1.** Publication-year accuracy in the release. The overall estimate is weighted; period rows are unweighted.

| Period | Books judged | Accuracy |
|---|---|---|
| Release (weighted) | 199 | 87.4% (95% CI 82.7–92.1) |
| 1700-99 | 16 | 81.2% |
| 1800-49 | 16 | 81.2% |
| 1850-99 | 67 | 89.6% |
| 1900-24 | 84 | 89.3% |
| 1925+ | 16 | 62.5% |

## 3. Dataset Description

**Table 2.** Construction, coverage, and quality profile of Gutenberg Atlas.

| Measure | Result |
|---|---|
| Eligible books | 38,550 |
| Released books | 21,296 (55.2% of eligible) |
| Distinct linked authors | 7,383 |
| Publication years | 1700–2004 (median 1901) |
| Gender coverage | 21,194 (99.5% of released) |
| Citizenship coverage | 18,978 (89.1% of released) |
| Flagged lifespan conflicts | 750 (3.5% of released) |
| Flagged dates | 7,509 (35.3% of released) |

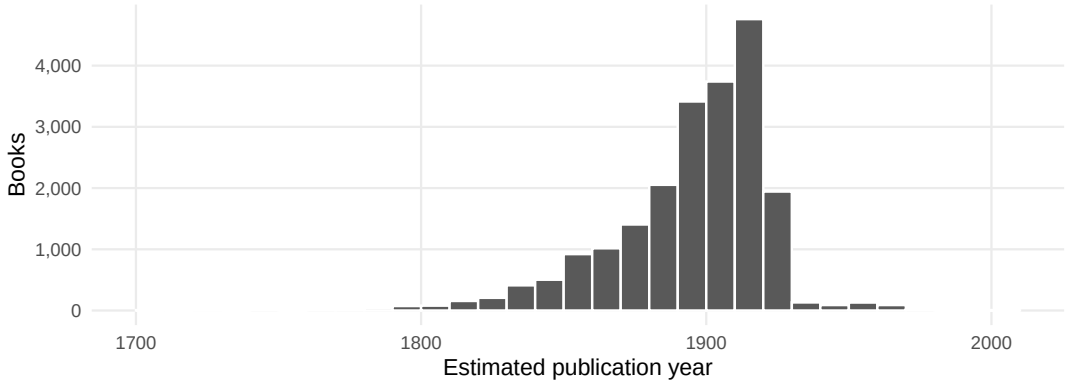



**Figure 1.** Estimated publication years of the 21,296 released books, by decade.

**Repository name:** Gutenberg Atlas (GitHub `jaytimm/gutenberg-atlas`, release v1.0.0)
**Object names:** `release.csv`, `unverified_candidates.csv`, generated summaries, validation files, build code, and documentation
**Formats:** UTF-8 CSV (data); Markdown (documentation); R and Python scripts (build code)
**Creation dates:** 2026-10-20 to 2026-10-22
**Dataset creator:** Jason Timm (conceptualization, methodology, software, validation, curation, analysis, and documentation)
**Language:** English variable names and documentation; books selected as English in the Project Gutenberg catalog
**License:** CC BY 4.0 (data); MIT (code)
**Publication date:** 2026-10-24

The canonical `release.csv` has one row per Project Gutenberg identifier. It records Gutenberg title and author fields; Wikidata identity, gender, citizenship, and lifespan claims; mapped present-day citizenship; HathiTrust year, item, imprint, match tier, and candidate count; and lifespan and date flags. The data dictionary separates source claims, derived fields, and validation results. Gender and citizenship retain Wikidata labels rather than being inferred from names or texts. The present-day citizenship mapping is a separate derived field, leaving users free to retain the historical claim or apply a different geographic scheme.

Rows represent ebooks, not unique works. In total, 3,055 rows share a HathiTrust item with another row (930 items). Work-level studies should deduplicate on `htid` or title and author.

The date flag marks uncertain imprints, multi-volume or collected-works titles, a single HathiTrust candidate, or more than 20 candidates; it describes rather than excludes. Unweighted validation accuracy was 80% for flagged and 90% for unflagged books. These flags permit sensitivity analyses without silently replacing or correcting the source-derived values.

`SOURCE_VERSIONS.md` documents snapshots and checksums; `RELEASE_FREEZE.md` identifies published artifacts. HathiTrust permits Hathifile reuse subject to contributing-institution restrictions (HathiTrust Digital Library, 2026b), and Wikidata is CC0. The 1.2 GB filtered HathiTrust extract is not redistributed, but frozen match results permit rebuilding the release without it.

# 4. Reuse Potential

Gutenberg Atlas supplies temporal and author covariates for historical corpus construction, stylometry, cultural analytics, book history, and teaching. Project Gutenberg identifiers support text retrieval under title-level terms and join directly to the Standardized Project Gutenberg Corpus (Gerlach & Font-Clos, 2020).

## 4.1 Worked reuse demonstration

As a reuse demonstration, models predicted broad genre, 25-year period, and continuous year in five author-disjoint folds. Genre used 480 books; temporal tasks used 1800–1924, an analytical window chosen for historical balance, not a release boundary. Models received up to 5,000 body-text words. Penalized models used word or character TF–IDF (Friedman et al., 2010); a semantic model averaged `BAAI/bge-base-en-v1.5` chunk embeddings (Xiao et al., 2023).

**Table 3.** Author-disjoint reuse demonstration. Sample gives books (authors). Lower mean absolute error (MAE) is better; higher macro-F1 is better.

| Task | Sample | Best body model | Baseline | Body-text result |
|---|---|---|---|---|
| Broad genre | 480 (362) | BGE embeddings | 0.189 macro-F1 | 0.865 macro-F1 |
| Publication period | 360 (268) | BGE embeddings | 0.308 macro-F1 | 0.441 macro-F1 |
| Publication year | 400 (301) | Word TF-IDF | 30.9 years MAE | 20.2 years MAE |

Body text substantially improved genre prediction, whereas the best model missed year by about 20 years on average. Raw file openings predicted time better because front matter often prints dates; temporal-language studies should therefore strip it. Supplementary File 1 gives full specifications, controls, intervals, and confusion matrices. The exercise is not offered as a benchmark leaderboard. It demonstrates two practical uses of the release: constructing author-disjoint historical samples and testing whether apparent temporal prediction comes from linguistic signal or printed metadata. Keeping all books by an author in one fold also prevents the evaluation from rewarding author recognition across train and test sets. The strong genre result shows that the linked texts support controlled supervised tasks; the weaker temporal result cautions that useful date covariates need not be readily recoverable from body language alone.

### 4.2 Limitations

Project Gutenberg reflects public-domain law, volunteer selection, and preservation, not historical publishing. HathiTrust coverage is uneven and earliest matches can be reprints, biasing estimated years and publication ages later. Lifespan screening also truncates those errors. Wikidata claims are incomplete, especially for citizenship, and were not independently biographically validated. Provenance should be retained, and archive composition should not be treated as evidence of historical authorship or publication rates.

## Competing interests

The author has no competing interests to declare.

## Author roles

Jason Timm: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, and Writing – review and editing.

## AI declaration

Anthropic Claude Code assisted with manuscript drafting, code debugging, and bibliographic searches for the validation sample. Every reference year in the validation sample rests on a cited source recorded in `validation_sourced.csv`, which the author checked. The author revised and verified the resulting text, analyses, citations, and reported values and takes responsibility for the final content.

## Supplementary Files

**Supplementary File 1.** *Gutenberg Atlas: Supplementary Materials.* Corpus construction, composition, publication-year validation, author-linkage audit, and an author-disjoint text-modeling reuse demonstration (`supplement.pdf`).

## References


Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, *33*(1), 1–22. https://doi.org/10.18637/jss.v033.i01

Gerlach, M., & Font-Clos, F. (2020). A standardized Project Gutenberg corpus for statistical analysis of natural language and quantitative linguistics. *Entropy*, *22*(1), 126. https://doi.org/10.3390/e22010126

HathiTrust Digital Library. (2026a). *Hathifiles: Bibliographic metadata extracts*. https://www.hathitrust.org/member-libraries/resources-for-librarians/data-resources/hathifiles/

HathiTrust Digital Library. (2026b). *Metadata sharing and use policy*. https://www.hathitrust.org/the-collection/terms-conditions/metadata-sharing-and-use-policy/

Project Gutenberg. (n.d.). *Offline catalogs: Project gutenberg catalog metadata in machine-readable format*. Retrieved October 20, 2026, from https://www.gutenberg.org/ebooks/offline_catalogs.html

Rae, J. W., Potapenko, A., Jayakumar, S. M., Hillier, C., & Lillicrap, T. P. (2019). Compressive transformers for long-range sequence modelling. *arXiv*. https://doi.org/10.48550/arXiv.1911.05507

Rohrbacher, K. (2025). De-corp: A corpus of german-language fiction and non-fiction (1780–1930). *Journal of Open Humanities Data*, *11*. https://doi.org/10.5334/johd.350

Rohrbacher, K., & Schrittesser, D. (2026). Bridging the gaps: Integrating bibliographic metadata into Wikidata for literary corpora. *Journal of Open Humanities Data*, *12*, 37. https://doi.org/10.5334/johd.483

Timm, J. (2026). *Gutenberg atlas: Publication-year and author metadata for english-language project gutenberg books* (Version 1.0.0). Zenodo. https://doi.org/10.5281/zenodo.23628668

Underwood, T., Kimutis, P., & Witte, J. (2020). NovelTM datasets for english-language fiction, 1700–2009. *Journal of Cultural Analytics*, *5*(2). https://doi.org/10.22148/001c.13147

Vrandečić, D., & Krötzsch, M. (2014). Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, *57*(10), 78–85. https://doi.org/10.1145/2629489

Xiao, S., Liu, Z., Zhang, P., & Muennighoff, N. (2023). *C-Pack: Packaged resources to advance general chinese embedding*. arXiv. https://doi.org/10.48550/arXiv.2309.07597